AI ASSISTED CODING
ASSIGNMENT - 4.1
ROLL NO. 2303A510C8

Task 1 – Portfolio Website Design
You are building a personal portfolio website to showcase your work.
Requirements:
• Create sections for About Me, Projects, and Contact.
• Use AI to:
o Suggest color palettes and typography.
o Create a responsive layout with Grid/Flexbox.
o Add smooth scrolling navigation.

Prompt:
I am creating a simple personal portfolio website using HTML, CSS and a little JavaScript.

Please help me build a responsive webpage that includes the following sections:

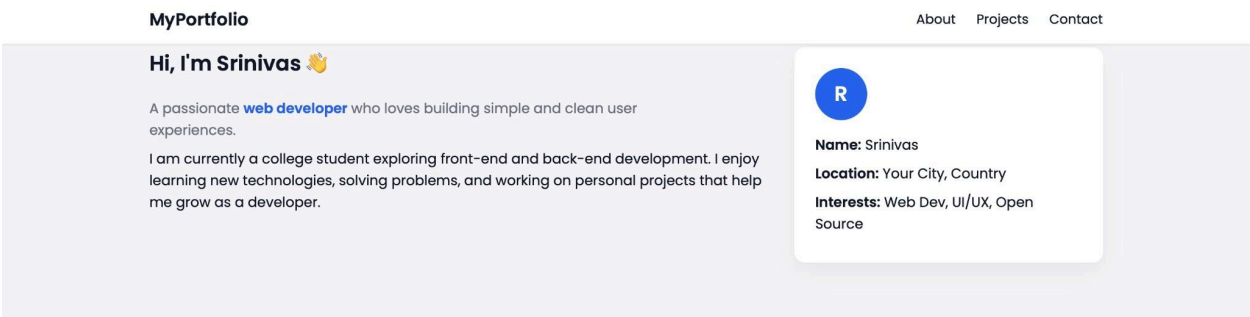
*** About Me**
*** Projects**
*** Contact**

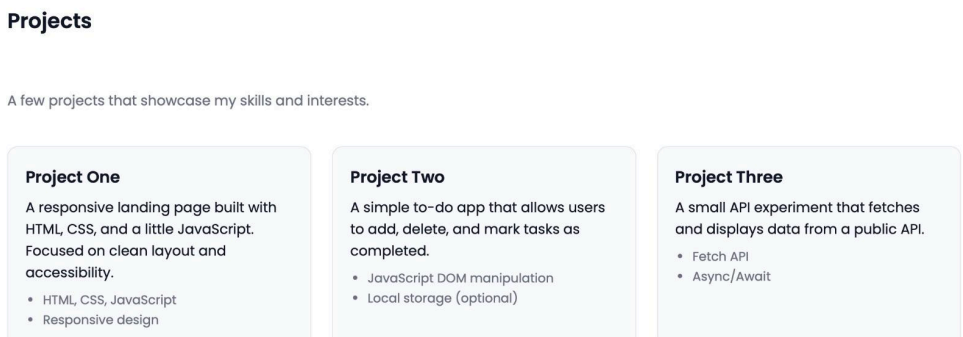
Requirements:

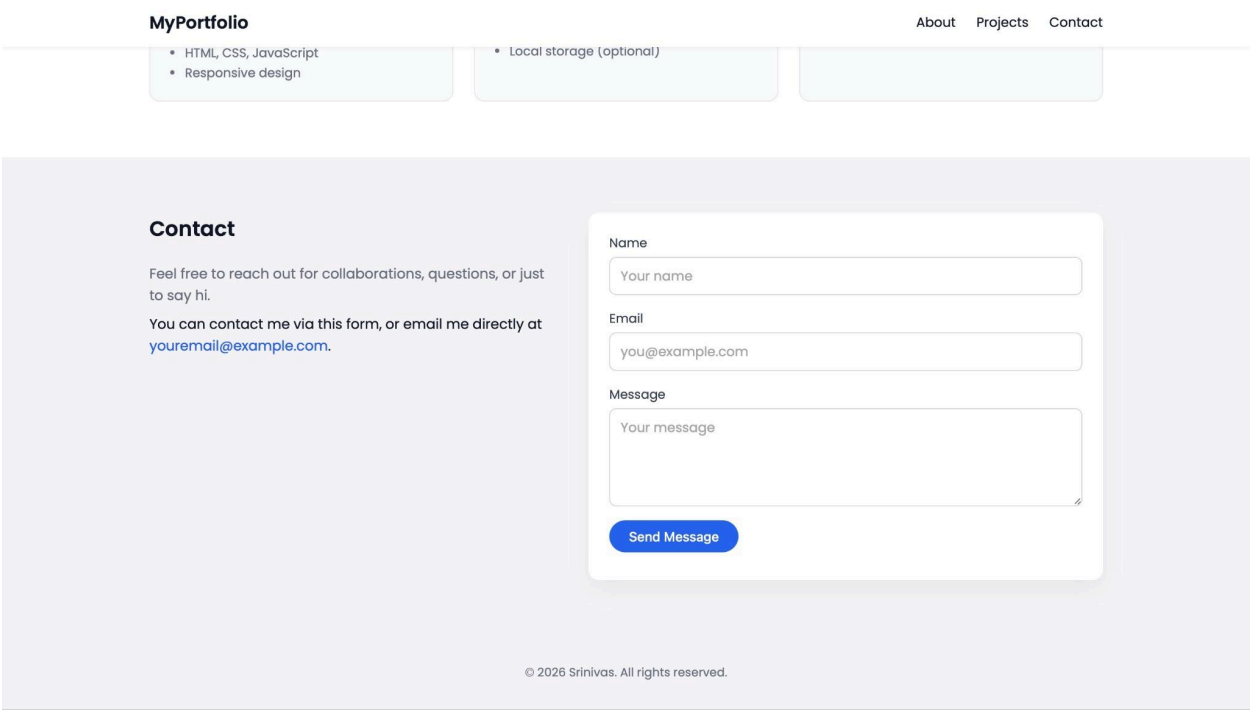
1. Suggest a simple color palette and good typography for a portfolio site.
2. Create a responsive layout using CSS Flexbox or Grid.
3. Add a navigation bar that scrolls smoothly to each section.
4. Keep the code clean and easy to understand since this is for a college assignment.

5. Provide HTML, CSS, and JavaScript in separate sections with comments explaining important parts.

OUTPUT:







Task 2 – Online Store Product Page
Design a product display page for an online store.
Requirements:
• Display product image, title, price, and "Add to Cart" button.
• Use AI to:
o Style with BEM methodology.
o Make layout responsive.
o Add hover effects and "Add to Cart" alert.
Prompt:
I am designing a simple product page for an online store using HTML, CSS, and JavaScript.

The page should display:

*** Product image**
*** Product title**
*** Product price**

* "Add to Cart" button

Requirements:

1. Use the BEM naming methodology for CSS classes.
2. Make the layout responsive for mobile and desktop.
3. Add hover effects for the product card and button.
4. When the "Add to Cart" button is clicked, show a JavaScript alert saying "Product added to cart".
5. Keep the code simple and well commented so it is easy to understand.

CODE:

```
<!DOCTYPE html>
<html lang="en">
<head>
 <meta charset="UTF-8" />
 <title>Simple Product Page</title>
 <meta name="viewport" content="width=device-width, initial-scale=1.0" />

 <style>
   /* ========== Page Base Styles ========== */
   * {
     box-sizing: border-box;
     margin: 0;
     padding: 0;
   }

   body {
     font-family: Arial, Helvetica, sans-serif;
     background-color: #f5f5f7;
     color: #222;
     display: flex;
     justify-content: center;
     align-items: center;
     min-height: 100vh;
     padding: 16px;
```

```
}

/* ========== Product Card (BEM: product-card) ========== */
.product-card {
  background-color: #ffffff;
  border-radius: 12px;
  box-shadow: 0 10px 25px rgba(0, 0, 0, 0.08);
  max-width: 360px; /* desktop width */
  width: 100%;
  overflow: hidden;
  transition: transform 0.2s ease, box-shadow 0.2s ease;
}

/* Card hover effect */
.product-card:hover {
  transform: translateY(-4px);
  box-shadow: 0 14px 30px rgba(0, 0, 0, 0.12);
}

/* ========== Product Image ========== */
.product-card__image-wrapper {
  width: 100%;
  overflow: hidden;
}

.product-card__image {
  display: block;
  width: 100%;
  height: auto;
  object-fit: cover;
}

/* ========== Product Content (title, price, button) ========== */
.product-card__content {
  padding: 16px 18px 20px;
}

.product-card__title {
  font-size: 1.25rem;
  font-weight: 600;
  margin-bottom: 8px;
}
```

```
.product-card__price {
  font-size: 1.1rem;
  font-weight: 700;
  color: #1a7b3f; /* green for price */
  margin-bottom: 16px;
}

/* ========== Add to Cart Button ========== */
.product-card__button {
  display: inline-block;
  width: 100%;
  text-align: center;
  padding: 10px 16px;
  background-color: #0070f3;
  color: #ffffff;
  border: none;
  border-radius: 8px;
  font-size: 1rem;
  font-weight: 600;
  cursor: pointer;
  transition: background-color 0.2s ease, transform 0.2s ease,
    box-shadow 0.2s ease;
}

/* Button hover effect */
.product-card__button:hover {
  background-color: #0056c2;
  transform: translateY(-1px);
  box-shadow: 0 6px 14px rgba(0, 0, 0, 0.18);
}

.product-card__button:active {
  transform: translateY(0);
  box-shadow: none;
}

/* ========== Responsive Adjustments ========== */
/*
  On very small screens, we already use full width with padding around body,
  so no extra work is needed. Below is an example breakpoint for larger
  screens if you want to tweak spacing.
```

```
  */
  @media (min-width: 768px) {
    body {
      padding: 32px;
    }
  }
</style>
</head>
<body>
<!--
  Product card container.
  BEM block: product-card
-->
<div class="product-card">
  <!-- Image section -->
  <div class="product-card__image-wrapper">
    <!--
      Replace src with your own image path if you have one locally.
      This placeholder uses an online image.
    -->
    <img
      class="product-card__image"
      src="https://via.placeholder.com/600x400?text=Product+Image"
      alt="Sample product image"
    />
  </div>

  <!-- Text and button section -->
  <div class="product-card__content">
    <h1 class="product-card__title">Wireless Headphones</h1>
    <p class="product-card__price">$79.99</p>

    <!-- Add to cart button (BEM element: product-card__button) -->
    <button class="product-card__button" id="add-to-cart-button">
      Add to Cart
    </button>
  </div>
</div>

<script>
  //
  // When the "Add to Cart" button is clicked,
```

```
  // show an alert saying "Product added to cart".
  //
  const addToCartButton = document.getElementById("add-to-cart-button");

  addToCartButton.addEventListener("click", function () {
    alert("Product added to cart");
  });
 </script>
</body>
</html><!DOCTYPE html>
<html lang="en">
<head>
 <meta charset="UTF-8" />
 <title>Simple Product Page</title>
 <meta name="viewport" content="width=device-width, initial-scale=1.0" />

 <style>
  /* ========== Page Base Styles ========== */
  * {
    box-sizing: border-box;
    margin: 0;
    padding: 0;
  }

  body {
    font-family: Arial, Helvetica, sans-serif;
    background-color: #f5f5f7;
    color: #222;
    display: flex;
    justify-content: center;
    align-items: center;
    min-height: 100vh;
    padding: 16px;
  }

  /* ========== Product Card (BEM: product-card) ========== */
  .product-card {
    background-color: #ffffff;
    border-radius: 12px;
    box-shadow: 0 10px 25px rgba(0, 0, 0, 0.08);
    max-width: 360px; /* desktop width */
    width: 100%;
```

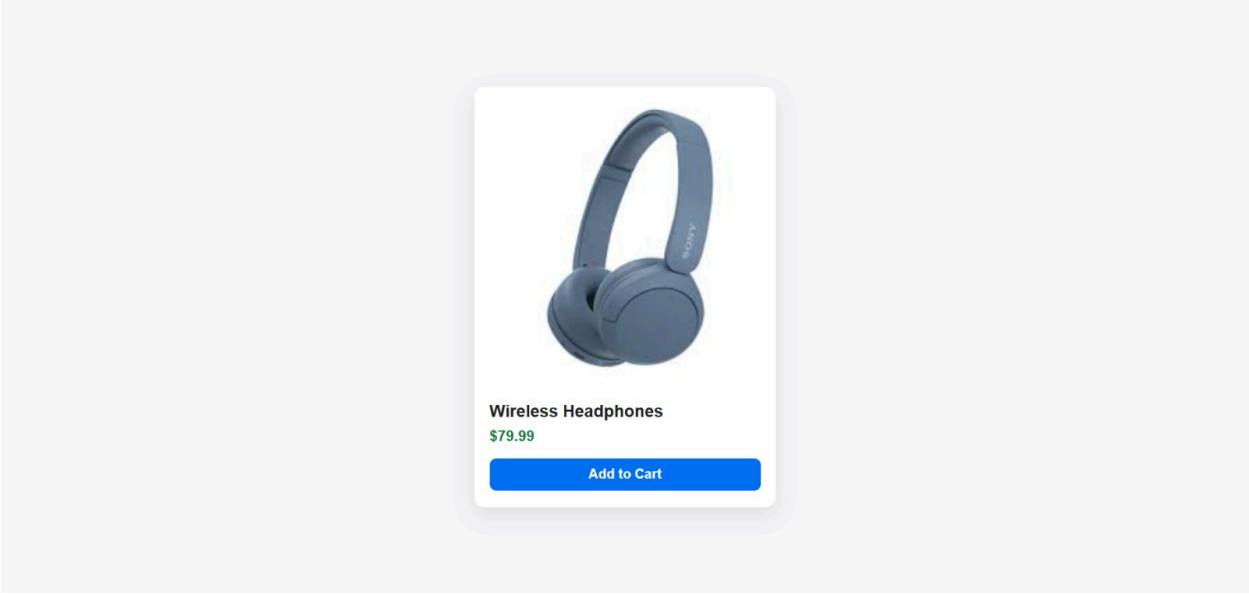
```
  overflow: hidden;
  transition: transform 0.2s ease, box-shadow 0.2s ease;
}

/* Card hover effect */
.product-card:hover {
  transform: translateY(-4px);
  box-shadow: 0 14px 30px rgba(0, 0, 0, 0.12);
}

/* ========== Product Image ========== */
.product-card__image-wrapper {
  width: 100%;
  overflow: hidden;
}

.product-card__image {
  display: block;
  width: 100%;
  height: auto;
  object-fit: cover;
}

/* ========== Product Content (title, price, button) ========== */
.product-card__content {
  padding: 16px 18px 20px;
}

.product-card__title {
  font-size: 1.25rem;
  font-weight: 600;
  margin-bottom: 8px;
}

.product-card__price {
  font-size: 1.1rem;
  font-weight: 700;
  color: #1a7b3f; /* green for price */
  margin-bottom: 16px;
}

/* ========== Add to Cart Button ========== */
```

```
  .product-card__button {
    display: inline-block;
    width: 100%;
    text-align: center;
    padding: 10px 16px;
    background-color: #0070f3;
    color: #ffffff;
    border: none;
    border-radius: 8px;
    font-size: 1rem;
    font-weight: 600;
    cursor: pointer;
    transition: background-color 0.2s ease, transform 0.2s ease,
      box-shadow 0.2s ease;
  }

  /* Button hover effect */
  .product-card__button:hover {
    background-color: #0056c2;
    transform: translateY(-1px);
    box-shadow: 0 6px 14px rgba(0, 0, 0, 0.18);
  }

  .product-card__button:active {
    transform: translateY(0);
    box-shadow: none;
  }

  /* ========== Responsive Adjustments ========== */
  /*
    On very small screens, we already use full width with padding around body,
    so no extra work is needed. Below is an example breakpoint for larger
    screens if you want to tweak spacing.
  */
  @media (min-width: 768px) {
    body {
      padding: 32px;
    }
  }
 </style>
</head>
<body>
```

```html
<!--
  Product card container.
  BEM block: product-card
-->
<div class="product-card">
  <!-- Image section -->
  <div class="product-card__image-wrapper">
    <!--
      Replace src with your own image path if you have one locally.
      This placeholder uses an online image.
    -->
    <img
      class="product-card__image"
      src="https://via.placeholder.com/600x400?text=Product+Image"
      alt="Sample product image"
    />
  </div>

  <!-- Text and button section -->
  <div class="product-card__content">
    <h1 class="product-card__title">Wireless Headphones</h1>
    <p class="product-card__price">$79.99</p>

    <!-- Add to cart button (BEM element: product-card__button) -->
    <button class="product-card__button" id="add-to-cart-button">
      Add to Cart
    </button>
  </div>
</div>

<script>
  //
  // When the "Add to Cart" button is clicked,
  // show an alert saying "Product added to cart".
  //
  const addToCartButton = document.getElementById("add-to-cart-button");

  addToCartButton.addEventListener("click", function () {
    alert("Product added to cart");
  });
</script>
</body>
```

```
</html>
```

OUTPUT:



Task 3 – Event Registration Form
Build an event registration form for a conference.
Requirements:
• Collect name, email, phone number, and session selection.
• Use AI to:
o Add form validation with JavaScript.
o Make the form accessible with labels and ARIA.
o Style with a professional look.
Prompt:
I need to build an event registration form for a conference using HTML, CSS, and JavaScript.

The form should collect:

*** Name**
*** Email**

* Phone Number
* Session selection (dropdown)

Requirements:

1. Add JavaScript form validation for required fields and proper email format.
2. Make the form accessible using proper labels and ARIA attributes.
3. Style the form so it looks professional and clean.
4. Make sure the layout is responsive for smaller screens.
5. Include comments explaining the validation logic.

CODE:

```html
<!DOCTYPE html>
<html lang="en">
<head>
 <meta charset="UTF-8" />
 <title>Conference Event Registration</title>
 <meta name="viewport" content="width=device-width, initial-scale=1.0" />

 <style>
   /* ====== Base styles ====== */
   * {
     box-sizing: border-box;
   }

   body {
     margin: 0;
     font-family: system-ui, -apple-system, BlinkMacSystemFont, "Segoe UI",
       sans-serif;
     background: #f3f4f6;
     color: #111827;
     display: flex;
     justify-content: center;
     padding: 24px;
   }
```

```
.page-wrapper {
  max-width: 600px;
  width: 100%;
}

.card {
  background: #ffffff;
  border-radius: 12px;
  padding: 24px 24px 28px;
  box-shadow: 0 10px 30px rgba(15, 23, 42, 0.12);
}

h1 {
  margin-top: 0;
  font-size: 1.6rem;
  margin-bottom: 4px;
}

.subtitle {
  margin: 0 0 20px;
  color: #6b7280;
  font-size: 0.95rem;
}

/* ====== Form layout ====== */
form {
  margin-top: 8px;
}

.form-group {
  margin-bottom: 16px;
}

label {
  display: inline-block;
  margin-bottom: 6px;
  font-weight: 600;
  font-size: 0.95rem;
  color: #111827;
}

.required-badge {
```

```
  color: #dc2626;
  margin-left: 4px;
  font-weight: 600;
}

input[type="text"],
input[type="email"],
input[type="tel"],
select {
  width: 100%;
  padding: 10px 12px;
  border-radius: 8px;
  border: 1px solid #d1d5db;
  font-size: 0.95rem;
  transition: border-color 0.15s ease, box-shadow 0.15s ease,
    background-color 0.15s ease;
  background-color: #f9fafb;
}

input:focus,
select:focus {
  outline: none;
  border-color: #2563eb;
  box-shadow: 0 0 0 3px rgba(37, 99, 235, 0.25);
  background-color: #ffffff;
}

.error {
  border-color: #dc2626 !important;
  background-color: #fef2f2;
}

.error-message {
  margin-top: 4px;
  font-size: 0.82rem;
  color: #b91c1c;
  min-height: 1.1em;
}

.form-footer {
  margin-top: 18px;
  display: flex;
```

```
  flex-direction: column;
  gap: 8px;
}

.submit-btn {
  display: inline-flex;
  align-items: center;
  justify-content: center;
  gap: 6px;
  padding: 10px 16px;
  border-radius: 999px;
  border: none;
  background: linear-gradient(135deg, #2563eb, #1d4ed8);
  color: #ffffff;
  font-weight: 600;
  font-size: 0.95rem;
  cursor: pointer;
  box-shadow: 0 10px 18px rgba(37, 99, 235, 0.35);
  transition: transform 0.1s ease, box-shadow 0.1s ease,
    background 0.15s ease;
}

.submit-btn:hover {
  transform: translateY(-1px);
  box-shadow: 0 12px 22px rgba(37, 99, 235, 0.4);
  background: linear-gradient(135deg, #1d4ed8, #1d4ed8);
}

.submit-btn:active {
  transform: translateY(0);
  box-shadow: 0 6px 10px rgba(37, 99, 235, 0.3);
}

.submit-btn:focus-visible {
  outline: 3px solid #1d4ed8;
  outline-offset: 3px;
}

.helper-text {
  font-size: 0.8rem;
  color: #6b7280;
}
```

```
  .success-banner {
    margin-top: 10px;
    padding: 8px 10px;
    border-radius: 8px;
    background-color: #ecfdf3;
    color: #166534;
    font-size: 0.88rem;
    border: 1px solid #bbf7d0;
    display: none;
  }

  .success-banner[aria-hidden="false"] {
    display: block;
  }

  /* ====== Responsive tweaks ====== */
  @media (max-width: 480px) {
    body {
      padding: 16px;
    }

    .card {
      padding: 18px 16px 24px;
    }

    h1 {
      font-size: 1.35rem;
    }
  }
 </style>
</head>
<body>
 <div class="page-wrapper">
  <section class="card" aria-labelledby="form-heading">
    <h1 id="form-heading">Conference Registration</h1>
    <p class="subtitle">
      Reserve your spot for the upcoming conference. Fields marked with
      <span class="required-badge" aria-hidden="true">*</span> are required.
    </p>

    <!-- ARIA live region for form-level status messages -->
```

```html
<div
  id="form-status"
  class="success-banner"
  role="status"
  aria-live="polite"
  aria-hidden="true"
></div>

<form id="registration-form" novalidate aria-describedby="form-help">
  <p id="form-help" class="helper-text">
    Your information is used only for event communication.
  </p>

  <!-- Name -->
  <div class="form-group">
    <label for="name">
      Full Name
      <span class="required-badge" aria-hidden="true">*</span>
    </label>
    <input
      type="text"
      id="name"
      name="name"
      autocomplete="name"
      aria-required="true"
      aria-describedby="name-error"
    />
    <div
      id="name-error"
      class="error-message"
      role="alert"
      aria-live="polite"
    ></div>
  </div>

  <!-- Email -->
  <div class="form-group">
    <label for="email">
      Email
      <span class="required-badge" aria-hidden="true">*</span>
    </label>
    <input
```

```
        type="email"
        id="email"
        name="email"
        autocomplete="email"
        inputmode="email"
        aria-required="true"
        aria-describedby="email-error"
      />
      <div
        id="email-error"
        class="error-message"
        role="alert"
        aria-live="polite"
      ></div>
    </div>

    <!-- Phone -->
    <div class="form-group">
      <label for="phone">
        Phone Number
        <span class="required-badge" aria-hidden="true">*</span>
      </label>
      <input
        type="tel"
        id="phone"
        name="phone"
        autocomplete="tel"
        inputmode="tel"
        aria-required="true"
        aria-describedby="phone-error"
      />
      <div
        id="phone-error"
        class="error-message"
        role="alert"
        aria-live="polite"
      ></div>
    </div>

    <!-- Session selection -->
    <div class="form-group">
      <label for="session">
```

```
          Session Selection
          <span class="required-badge" aria-hidden="true">*</span>
        </label>
        <select
          id="session"
          name="session"
          aria-required="true"
          aria-describedby="session-error"
        >
          <option value="">-- Please choose a session --</option>
          <option value="keynote">Keynote & Opening Remarks</option>
          <option value="track-ai">AI & Machine Learning Track</option>
          <option value="track-web">Modern Web Development Track</option>
          <option value="track-data">Data Science & Analytics Track</option>
          <option value="workshop">Hands-on Workshop (Afternoon)</option>
        </select>
        <div
          id="session-error"
          class="error-message"
          role="alert"
          aria-live="polite"
        ></div>
      </div>

      <div class="form-footer">
        <button type="submit" class="submit-btn">
          Submit Registration
        </button>
      </div>
    </form>
  </section>
</div>

<script>
  // ====== Simple form validation logic ======
  // We validate on submit, and also clear errors when the user starts typing again.

  const form = document.getElementById("registration-form");
  const statusBanner = document.getElementById("form-status");

  // Helper to show an error on a field
  function showError(inputEl, errorEl, message) {
```

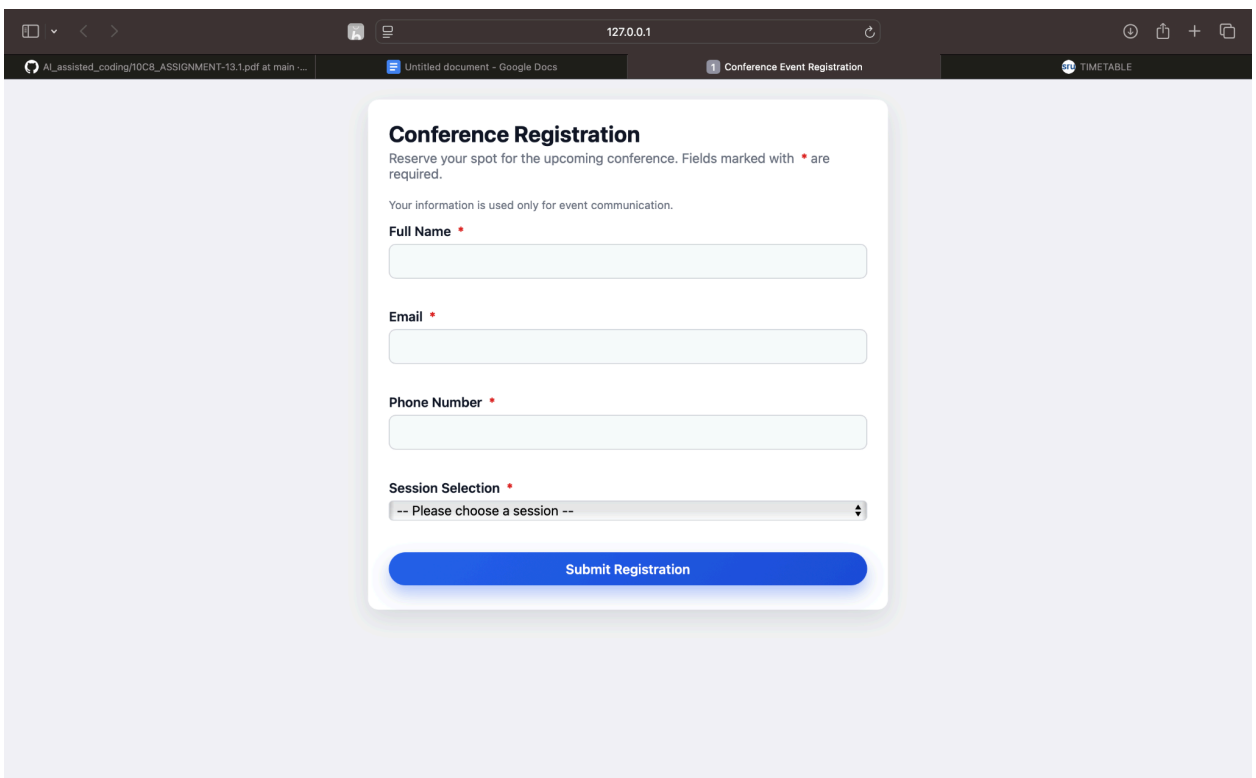
```
    inputEl.classList.add("error");
    inputEl.setAttribute("aria-invalid", "true");
    errorEl.textContent = message;
  }

  // Helper to clear an error on a field
  function clearError(inputEl, errorEl) {
    inputEl.classList.remove("error");
    inputEl.removeAttribute("aria-invalid");
    errorEl.textContent = "";
  }

  // Basic email pattern for client-side validation (not perfect, but good enough)
  function isValidEmail(email) {
    const emailPattern = /^[^\s@]+@[^\s@]+\.[^\s@]+$/;
    return emailPattern.test(email.trim());
  }

  // Basic phone validation: at least 7 digits, ignoring spaces, dashes, and
parentheses
  function isValidPhone(phone) {
    const digits = phone.replace(/\D/g, "");
    return digits.length >= 7;
  }

  form.addEventListener("submit", function (event) {
    event.preventDefault(); // prevent default form submission

    // Get references to inputs and error containers
    const nameInput = document.getElementById("name");
    const emailInput = document.getElementById("email");
    const phoneInput = document.getElementById("phone");
    const sessionSelect = document.getElementById("session");

    const nameError = document.getElementById("name-error");
    const emailError = document.getElementById("email-error");
    const phoneError = document.getElementById("phone-error");
    const sessionError = document.getElementById("session-error");

    // Clear previous status message
    statusBanner.textContent = "";
    statusBanner.setAttribute("aria-hidden", "true");
```

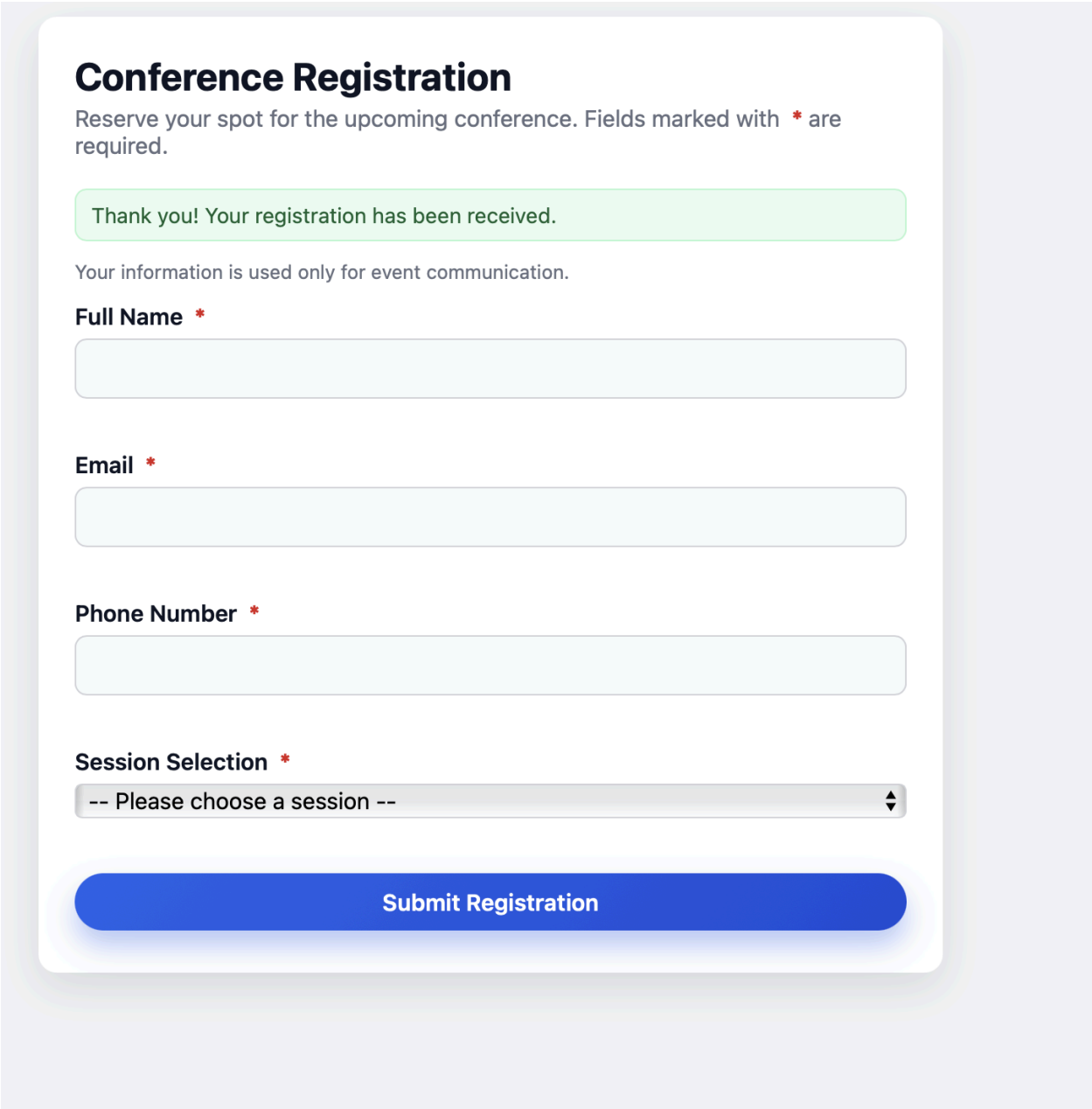
```
// Clear previous field errors
clearError(nameInput, nameError);
clearError(emailInput, emailError);
clearError(phoneInput, phoneError);
clearError(sessionSelect, sessionError);

let hasError = false;

// ==== Required field checks ====

// Name required
if (!nameInput.value.trim()) {
  showError(nameInput, nameError, "Please enter your full name.");
  hasError = true;
}

// Email required + format
if (!emailInput.value.trim()) {
  showError(emailInput, emailError, "Please enter your email address.");
  hasError = true;
} else if (!isValidEmail(emailInput.value)) {
  // Email format validation using simple regex pattern
  showError(
    emailInput,
    emailError,
    "Please enter a valid email address (e.g., name@example.com)."
  );
  hasError = true;
}

// Phone required + basic format
if (!phoneInput.value.trim()) {
  showError(phoneInput, phoneError, "Please enter your phone number.");
  hasError = true;
} else if (!isValidPhone(phoneInput.value)) {
  // Phone validation checks for at least 7 digits
  showError(
    phoneInput,
    phoneError,
    "Please enter a valid phone number with at least 7 digits."
  );
```

```
      hasError = true;
    }

    // Session selection required
    if (!sessionSelect.value) {
      showError(
        sessionSelect,
        sessionError,
        "Please select a session you would like to attend."
      );
      hasError = true;
    }

    // If any error exists, focus the first invalid field and stop
    if (hasError) {
      const firstErrorField = form.querySelector(
        ".error, [aria-invalid='true']"
      );
      if (firstErrorField) {
        firstErrorField.focus();
      }
      return;
    }

    // If no errors: simulate successful submission
    statusBanner.textContent =
      "Thank you! Your registration has been received.";
    statusBanner.setAttribute("aria-hidden", "false");

    // Optionally reset the form
    form.reset();
  });

  // Clear error message when user starts correcting a field
  ["input", "change"].forEach((eventType) => {
    form.addEventListener(
      eventType,
      function (event) {
        const target = event.target;
        if (!target.id) return;

        const errorEl = document.getElementById(target.id + "-error");
```

```
        if (errorEl) {
          clearError(target, errorEl);
        }
      },
      true
    );
  });
 </script>
</body>
</html>
```

OUTPUT:





Task 4 – Resume Website

Build a one-page online resume.

Requirements:

• Sections: Profile Summary, Skills, Education, Experience.

• Use AI to:

o Propose professional typography styles.

o Align sections using CSS Grid.

o Add smooth scroll navigation between sections.

Prompt:

I want to build a one-page resume website using HTML, CSS, and JavaScript.

The webpage should include these sections:

*** Profile Summary**
*** Skills**
*** Education**
*** Experience**

Requirements:

1. Suggest professional typography suitable for a resume website.
2. Align and structure sections using CSS Grid.
3. Add a simple navigation menu that smoothly scrolls to each section.
4. Keep the design clean and minimal like a professional resume.
5. Provide clear and commented code.

CODE:

```html
<!DOCTYPE html>
<html lang="en">

<head>
 <meta charset="UTF-8" />
 <meta name="viewport" content="width=device-width, initial-scale=1.0" />
 <title>Resume - Your Name</title>

 <!--
   Typography:
   - Uses "Segoe UI" (Windows), with fallbacks to system UI fonts.
```

```
  - Clean and professional, similar to what you see in MS Office.
-->
<style>
  /* ====== Base Reset & Layout ====== */
  *,
  *::before,
  *::after {
    box-sizing: border-box;
  }

  html {
    scroll-behavior: smooth;
  }

  body {
    margin: 0;
    padding: 0;
    font-family: -apple-system, BlinkMacSystemFont, "Segoe UI", Roboto,
      "Helvetica Neue", Arial, sans-serif;
    background-color: #f5f6f8;
    color: #222;
    line-height: 1.6;
  }

  /* ====== Page Wrapper ====== */
  .page {
    max-width: 960px;
    margin: 0 auto;
    padding: 24px 16px 40px;
  }

  /* ====== Header & Navigation ====== */
  header {
    position: sticky;
    top: 0;
    z-index: 10;
    background-color: #ffffffee;
    backdrop-filter: blur(6px);
    border-bottom: 1px solid #e0e0e0;
  }

  .header-inner {
```

```
  max-width: 960px;
  margin: 0 auto;
  padding: 12px 16px;
  display: flex;
  align-items: center;
  justify-content: space-between;
  gap: 16px;
}

.branding {
  display: flex;
  flex-direction: column;
}

.branding__name {
  font-size: 1.25rem;
  font-weight: 600;
  letter-spacing: 0.03em;
}

.branding__role {
  font-size: 0.9rem;
  color: #666;
}

/* Simple horizontal navigation */
nav ul {
  margin: 0;
  padding: 0;
  display: flex;
  list-style: none;
  gap: 16px;
  font-size: 0.9rem;
}

nav a {
  text-decoration: none;
  color: #444;
  padding: 4px 0;
  border-bottom: 2px solid transparent;
  transition: color 0.15s ease, border-color 0.15s ease;
}
```

```
nav a:hover,
nav a:focus-visible {
  color: #0057b8;
  border-color: #0057b8;
  outline: none;
}

/* Mobile nav: simple wrap */
@media (max-width: 640px) {
  .header-inner {
    flex-direction: column;
    align-items: flex-start;
  }

  nav ul {
    flex-wrap: wrap;
  }
}

/* ====== Main Layout Using CSS Grid ====== */
main {
  margin-top: 24px;
}

/*
  Use a single-column layout on small screens,
  and a two-column grid on larger screens.
*/
.grid {
  display: grid;
  grid-template-columns: 1fr;
  gap: 24px;
}

@media (min-width: 768px) {
  .grid {
    grid-template-columns: 2fr 3fr;
    align-items: start;
  }
}
```

```
/* ====== Card / Section Styling ====== */
section {
  background-color: #fff;
  border-radius: 6px;
  padding: 18px 20px;
  border: 1px solid #e0e0e0;
}

section + section {
  margin-top: 16px;
}

.section-title {
  font-size: 1.05rem;
  text-transform: uppercase;
  letter-spacing: 0.12em;
  color: #555;
  margin: 0 0 12px;
  border-bottom: 1px solid #eee;
  padding-bottom: 6px;
}

.muted {
  color: #777;
  font-size: 0.92rem;
}

/* ====== Profile Summary ====== */
#profile-summary p {
  margin-top: 4px;
  font-size: 0.98rem;
}

/* ====== Skills ====== */
.skills-list {
  display: grid;
  grid-template-columns: repeat(auto-fit, minmax(120px, 1fr));
  gap: 8px 16px;
  padding-left: 0;
  list-style: none;
  font-size: 0.95rem;
}
```

```
.skills-list li::before {
  content: "•";
  color: #0057b8;
  margin-right: 6px;
}

/* ====== Education & Experience ====== */
.entry {
  margin-bottom: 14px;
}

.entry:last-child {
  margin-bottom: 0;
}

.entry__header {
  display: flex;
  justify-content: space-between;
  gap: 12px;
  font-size: 0.96rem;
  font-weight: 500;
}

.entry__sub {
  font-size: 0.9rem;
  color: #666;
  margin-top: 2px;
}

.entry__details {
  margin-top: 4px;
  font-size: 0.92rem;
}

.entry__details ul {
  margin: 4px 0 0 16px;
  padding: 0;
}

.entry__details li {
  margin-bottom: 2px;
```

```
  }

  /* ====== Small Footer ====== */
  footer {
    margin-top: 32px;
    text-align: center;
    font-size: 0.85rem;
    color: #888;
  }
 </style>
</head>

<body>
 <!-- ====== Sticky Header with Navigation ====== -->
 <header>
   <div class="header-inner">
     <div class="branding">
       <span class="branding__name">Srinivas</span>
       <span class="branding__role">Job Title / Role</span>
     </div>
     <nav aria-label="Resume sections">
       <ul>
         <!--
           Each navigation link scrolls smoothly to its corresponding section.
           The href value must match the section's id.
         -->
         <li><a href="#profile-summary">Profile</a></li>
         <li><a href="#skills">Skills</a></li>
         <li><a href="#education">Education</a></li>
         <li><a href="#experience">Experience</a></li>
       </ul>
     </nav>
   </div>
 </header>

 <!-- ====== Main Resume Content ====== -->
 <div class="page">
   <main class="grid">
     <!-- Left column: Profile + Skills -->
     <div>
       <!-- Profile Summary Section -->
       <section id="profile-summary">
```

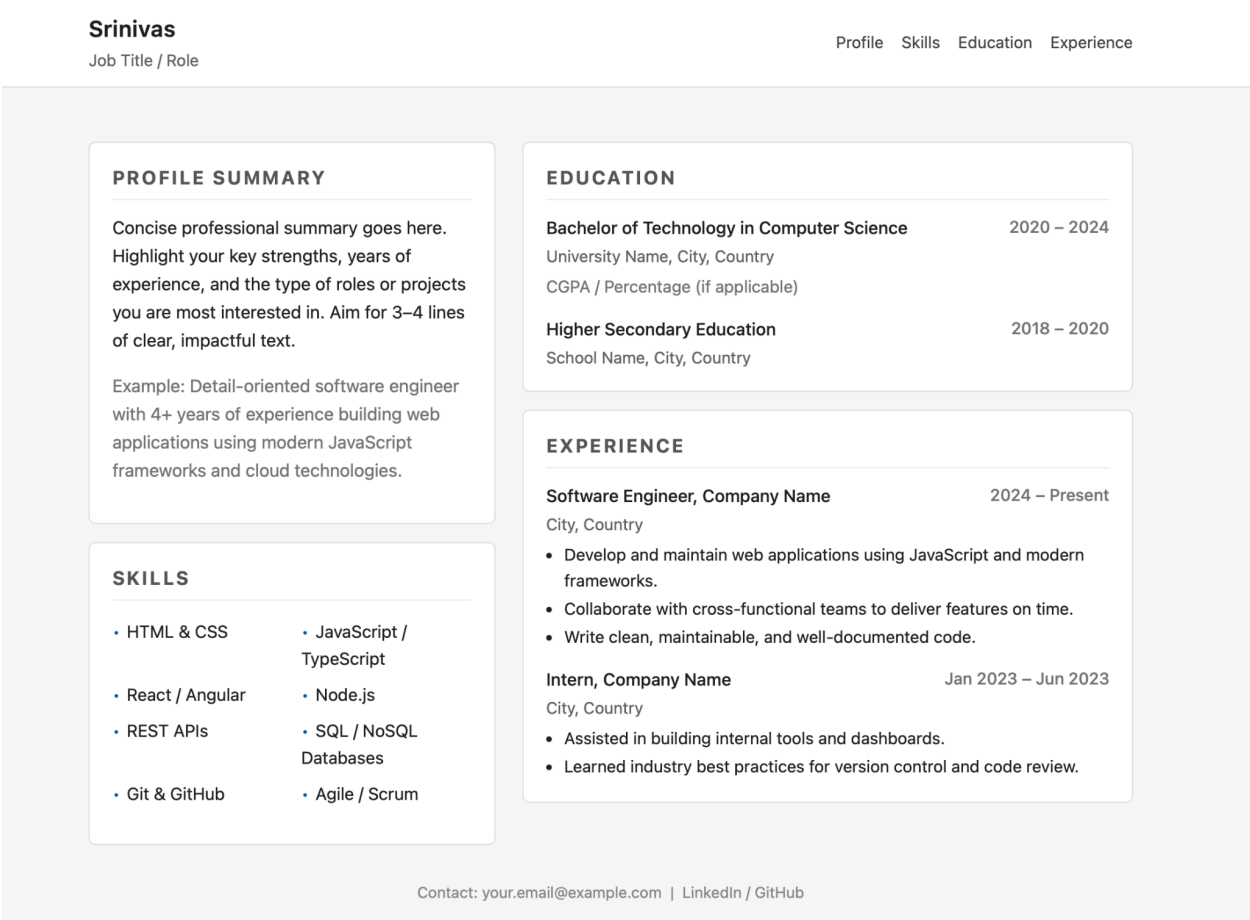
```html
      <h2 class="section-title">Profile Summary</h2>
      <p>
        Concise professional summary goes here. Highlight your key strengths,
        years of experience, and the type of roles or projects you are most
        interested in. Aim for 3–4 lines of clear, impactful text.
      </p>
      <p class="muted">
        Example: Detail-oriented software engineer with 4+ years of experience
        building web applications using modern JavaScript frameworks and cloud
        technologies.
      </p>
    </section>

    <!-- Skills Section -->
    <section id="skills">
      <h2 class="section-title">Skills</h2>
      <!--
        Skills are laid out using CSS Grid.
        Adjust the list items to match your actual skills.
      -->
      <ul class="skills-list">
        <li>HTML &amp; CSS</li>
        <li>JavaScript / TypeScript</li>
        <li>React / Angular</li>
        <li>Node.js</li>
        <li>REST APIs</li>
        <li>SQL / NoSQL Databases</li>
        <li>Git &amp; GitHub</li>
        <li>Agile / Scrum</li>
      </ul>
    </section>
  </div>

  <!-- Right column: Education + Experience -->
  <div>
    <!-- Education Section -->
    <section id="education">
      <h2 class="section-title">Education</h2>

      <div class="entry">
        <div class="entry__header">
          <span>Bachelor of Technology in Computer Science</span>
```

```
            <span class="muted">2020 – 2024</span>
          </div>
          <div class="entry__sub">
            University Name, City, Country
          </div>
          <div class="entry__details muted">
            CGPA / Percentage (if applicable)
          </div>
        </div>

        <div class="entry">
          <div class="entry__header">
            <span>Higher Secondary Education</span>
            <span class="muted">2018 – 2020</span>
          </div>
          <div class="entry__sub">
            School Name, City, Country
          </div>
        </div>
      </section>

      <!-- Experience Section -->
      <section id="experience">
        <h2 class="section-title">Experience</h2>

        <!-- Example job entry 1 -->
        <div class="entry">
          <div class="entry__header">
            <span>Software Engineer, Company Name</span>
            <span class="muted">2024 – Present</span>
          </div>
          <div class="entry__sub">
            City, Country
          </div>
          <div class="entry__details">
            <ul>
              <li>Develop and maintain web applications using JavaScript and modern
frameworks.</li>
              <li>Collaborate with cross-functional teams to deliver features on
time.</li>
              <li>Write clean, maintainable, and well-documented code.</li>
            </ul>
```

```
          </div>
        </div>

        <!-- Example job entry 2 -->
        <div class="entry">
          <div class="entry__header">
            <span>Intern, Company Name</span>
            <span class="muted">Jan 2023 – Jun 2023</span>
          </div>
          <div class="entry__sub">
            City, Country
          </div>
          <div class="entry__details">
            <ul>
              <li>Assisted in building internal tools and dashboards.</li>
              <li>Learned industry best practices for version control and code
review.</li>
            </ul>
          </div>
        </div>
      </section>
    </div>
  </main>

  <footer>
    <!--
      You can add contact information or links here,
      such as email, LinkedIn, or GitHub.
    -->
    <span>Contact: your.email@example.com  |  LinkedIn / GitHub</span>
  </footer>
</div>

<!-- ====== Minimal JavaScript for Nav Highlighting (Optional) ====== -->
<script>
  // This script highlights the nav link for the section currently in view.
  // It keeps the behavior simple and unobtrusive.

  document.addEventListener("DOMContentLoaded", function () {
    const navLinks = Array.from(document.querySelectorAll("nav a"));

    // Map section id to corresponding nav link
```

```
    const linkById = {};
    navLinks.forEach((link) => {
      const id = link.getAttribute("href").slice(1);
      linkById[id] = link;
    });

    const sections = Array.from(document.querySelectorAll("main section"));

    function onScroll() {
      let activeId = null;
      const offset = 120; // accounts for sticky header height

      sections.forEach((section) => {
        const rect = section.getBoundingClientRect();
        if (rect.top - offset <= 0 && rect.bottom - offset > 0) {
          activeId = section.id;
        }
      });

      navLinks.forEach((link) => {
        link.style.fontWeight = "400";
      });

      if (activeId && linkById[activeId]) {
        linkById[activeId].style.fontWeight = "600";
      }
    }

    document.addEventListener("scroll", onScroll, { passive: true });
    onScroll();
  });
 </script>
</body>

</html>
```

OUTPUT:



Task 5 – Restaurant Menu Page

Design a digital menu page for a restaurant.

Requirements:

• Display menu items with name, description, and price.

• Use AI to:

o Suggest a food-friendly color palette.

o Design a responsive card-based layout.

o Add hover effects on menu items.

Prompt:

I am designing a digital restaurant menu page using HTML and CSS.

Each menu item should show:

*** Item name**
*** Short description**
*** Price**

Requirements:

1. Suggest a color palette that fits a food or restaurant theme.
2. Display menu items using a responsive card-based layout.
3. Add hover effects on menu cards to make the design interactive.
4. Ensure the page works well on both mobile and desktop screens.
5. Keep the code simple and readable.

CODE:

```
<!DOCTYPE html>
<html lang="en">

<head>
 <meta charset="UTF-8" />
 <meta name="viewport" content="width=device-width, initial-scale=1.0" />
 <title>Resume - Your Name</title>

 <!--
   Typography:
   - Uses "Segoe UI" (Windows), with fallbacks to system UI fonts.
   - Clean and professional, similar to what you see in MS Office.
 -->
 <style>
   /* ====== Base Reset & Layout ====== */
   *,
   *::before,
   *::after {
```

```
  box-sizing: border-box;
}

html {
  scroll-behavior: smooth;
}

body {
  margin: 0;
  padding: 0;
  font-family: -apple-system, BlinkMacSystemFont, "Segoe UI", Roboto,
    "Helvetica Neue", Arial, sans-serif;
  background-color: #f5f6f8;
  color: #222;
  line-height: 1.6;
}

/* ====== Page Wrapper ====== */
.page {
  max-width: 960px;
  margin: 0 auto;
  padding: 24px 16px 40px;
}

/* ====== Header & Navigation ====== */
header {
  position: sticky;
  top: 0;
  z-index: 10;
  background-color: #ffffffee;
  backdrop-filter: blur(6px);
  border-bottom: 1px solid #e0e0e0;
}

.header-inner {
  max-width: 960px;
  margin: 0 auto;
  padding: 12px 16px;
  display: flex;
  align-items: center;
  justify-content: space-between;
  gap: 16px;
```

```
}

.branding {
  display: flex;
  flex-direction: column;
}

.branding__name {
  font-size: 1.25rem;
  font-weight: 600;
  letter-spacing: 0.03em;
}

.branding__role {
  font-size: 0.9rem;
  color: #666;
}

/* Simple horizontal navigation */
nav ul {
  margin: 0;
  padding: 0;
  display: flex;
  list-style: none;
  gap: 16px;
  font-size: 0.9rem;
}

nav a {
  text-decoration: none;
  color: #444;
  padding: 4px 0;
  border-bottom: 2px solid transparent;
  transition: color 0.15s ease, border-color 0.15s ease;
}

nav a:hover,
nav a:focus-visible {
  color: #0057b8;
  border-color: #0057b8;
  outline: none;
}
```

```css
/* Mobile nav: simple wrap */
@media (max-width: 640px) {
  .header-inner {
    flex-direction: column;
    align-items: flex-start;
  }

  nav ul {
    flex-wrap: wrap;
  }
}

/* ====== Main Layout Using CSS Grid ====== */
main {
  margin-top: 24px;
}

/*
  Use a single-column layout on small screens,
  and a two-column grid on larger screens.
*/
.grid {
  display: grid;
  grid-template-columns: 1fr;
  gap: 24px;
}

@media (min-width: 768px) {
  .grid {
    grid-template-columns: 2fr 3fr;
    align-items: start;
  }
}

/* ====== Card / Section Styling ====== */
section {
  background-color: #fff;
  border-radius: 6px;
  padding: 18px 20px;
  border: 1px solid #e0e0e0;
}
```

```
section + section {
  margin-top: 16px;
}

.section-title {
  font-size: 1.05rem;
  text-transform: uppercase;
  letter-spacing: 0.12em;
  color: #555;
  margin: 0 0 12px;
  border-bottom: 1px solid #eee;
  padding-bottom: 6px;
}

.muted {
  color: #777;
  font-size: 0.92rem;
}

/* ====== Profile Summary ====== */
#profile-summary p {
  margin-top: 4px;
  font-size: 0.98rem;
}

/* ====== Skills ====== */
.skills-list {
  display: grid;
  grid-template-columns: repeat(auto-fit, minmax(120px, 1fr));
  gap: 8px 16px;
  padding-left: 0;
  list-style: none;
  font-size: 0.95rem;
}

.skills-list li::before {
  content: "•";
  color: #0057b8;
  margin-right: 6px;
}
```

```
/* ====== Education & Experience ====== */
.entry {
  margin-bottom: 14px;
}

.entry:last-child {
  margin-bottom: 0;
}

.entry__header {
  display: flex;
  justify-content: space-between;
  gap: 12px;
  font-size: 0.96rem;
  font-weight: 500;
}

.entry__sub {
  font-size: 0.9rem;
  color: #666;
  margin-top: 2px;
}

.entry__details {
  margin-top: 4px;
  font-size: 0.92rem;
}

.entry__details ul {
  margin: 4px 0 0 16px;
  padding: 0;
}

.entry__details li {
  margin-bottom: 2px;
}

/* ====== Small Footer ====== */
footer {
  margin-top: 32px;
  text-align: center;
  font-size: 0.85rem;
```

```
    color: #888;
  }
</style>
</head>

<body>
<!-- ====== Sticky Header with Navigation ====== -->
<header>
  <div class="header-inner">
    <div class="branding">
      <span class="branding__name">Srinivas</span>
      <span class="branding__role">Job Title / Role</span>
    </div>
    <nav aria-label="Resume sections">
      <ul>
        <!--
          Each navigation link scrolls smoothly to its corresponding section.
          The href value must match the section's id.
        -->
        <li><a href="#profile-summary">Profile</a></li>
        <li><a href="#skills">Skills</a></li>
        <li><a href="#education">Education</a></li>
        <li><a href="#experience">Experience</a></li>
      </ul>
    </nav>
  </div>
</header>

<!-- ====== Main Resume Content ====== -->
<div class="page">
  <main class="grid">
    <!-- Left column: Profile + Skills -->
    <div>
      <!-- Profile Summary Section -->
      <section id="profile-summary">
        <h2 class="section-title">Profile Summary</h2>
        <p>
          Concise professional summary goes here. Highlight your key strengths,
          years of experience, and the type of roles or projects you are most
          interested in. Aim for 3-4 lines of clear, impactful text.
        </p>
        <p class="muted">
```

```
        Example: Detail-oriented software engineer with 4+ years of experience
        building web applications using modern JavaScript frameworks and cloud
        technologies.
      </p>
    </section>

    <!-- Skills Section -->
    <section id="skills">
      <h2 class="section-title">Skills</h2>
      <!--
        Skills are laid out using CSS Grid.
        Adjust the list items to match your actual skills.
      -->
      <ul class="skills-list">
        <li>HTML &amp; CSS</li>
        <li>JavaScript / TypeScript</li>
        <li>React / Angular</li>
        <li>Node.js</li>
        <li>REST APIs</li>
        <li>SQL / NoSQL Databases</li>
        <li>Git &amp; GitHub</li>
        <li>Agile / Scrum</li>
      </ul>
    </section>
  </div>

  <!-- Right column: Education + Experience -->
  <div>
    <!-- Education Section -->
    <section id="education">
      <h2 class="section-title">Education</h2>

      <div class="entry">
        <div class="entry__header">
          <span>Bachelor of Technology in Computer Science</span>
          <span class="muted">2020 – 2024</span>
        </div>
        <div class="entry__sub">
          University Name, City, Country
        </div>
        <div class="entry__details muted">
          CGPA / Percentage (if applicable)
```

```
          </div>
        </div>

        <div class="entry">
          <div class="entry__header">
            <span>Higher Secondary Education</span>
            <span class="muted">2018 – 2020</span>
          </div>
          <div class="entry__sub">
            School Name, City, Country
          </div>
        </div>
      </section>

      <!-- Experience Section -->
      <section id="experience">
        <h2 class="section-title">Experience</h2>

        <!-- Example job entry 1 -->
        <div class="entry">
          <div class="entry__header">
            <span>Software Engineer, Company Name</span>
            <span class="muted">2024 – Present</span>
          </div>
          <div class="entry__sub">
            City, Country
          </div>
          <div class="entry__details">
            <ul>
              <li>Develop and maintain web applications using JavaScript and modern
frameworks.</li>
              <li>Collaborate with cross-functional teams to deliver features on
time.</li>
              <li>Write clean, maintainable, and well-documented code.</li>
            </ul>
          </div>
        </div>

        <!-- Example job entry 2 -->
        <div class="entry">
          <div class="entry__header">
            <span>Intern, Company Name</span>
```

```
            <span class="muted">Jan 2023 – Jun 2023</span>
          </div>
          <div class="entry__sub">
            City, Country
          </div>
          <div class="entry__details">
            <ul>
              <li>Assisted in building internal tools and dashboards.</li>
              <li>Learned industry best practices for version control and code
review.</li>
            </ul>
          </div>
        </div>
      </section>
    </div>
  </main>

  <footer>
    <!--
      You can add contact information or links here,
      such as email, LinkedIn, or GitHub.
    -->
    <span>Contact: your.email@example.com  |  LinkedIn / GitHub</span>
  </footer>
</div>

<!-- ====== Minimal JavaScript for Nav Highlighting (Optional) ====== -->
<script>
  // This script highlights the nav link for the section currently in view.
  // It keeps the behavior simple and unobtrusive.

  document.addEventListener("DOMContentLoaded", function () {
    const navLinks = Array.from(document.querySelectorAll("nav a"));

    // Map section id to corresponding nav link
    const linkById = {};
    navLinks.forEach((link) => {
      const id = link.getAttribute("href").slice(1);
      linkById[id] = link;
    });

    const sections = Array.from(document.querySelectorAll("main section"));
```

```
    function onScroll() {
      let activeId = null;
      const offset = 120; // accounts for sticky header height

      sections.forEach((section) => {
        const rect = section.getBoundingClientRect();
        if (rect.top - offset <= 0 && rect.bottom - offset > 0) {
          activeId = section.id;
        }
      });

      navLinks.forEach((link) => {
        link.style.fontWeight = "400";
      });

      if (activeId && linkById[activeId]) {
        linkById[activeId].style.fontWeight = "600";
      }
    }

    document.addEventListener("scroll", onScroll, { passive: true });
    onScroll();
  });
 </script>
</body>

</html>
```

OUTPUT:

127.0.0.1

AI_assisted_coding/10C8_ASSIGNMENT-13.1.pdf at main ·... | Untitled document - Google Docs | Restaurant Menu | TIMETABLE

SUNSET BISTRO

Freshly prepared dishes made with seasonal ingredients.

STARTERS

Bruschetta Trio — $8.50

Toasted baguette topped with tomato basil, roasted peppers, and whipped goat cheese.

Creamy Tomato Soup — $6.00

Slow-simmered tomatoes with basil, topped with herb croutons.

MAINS

Grilled Lemon Herb Chicken — $16.90

Marinated chicken breast with roasted potatoes and seasonal vegetables.

Wild Mushroom Risotto — $15.50

Arborio rice cooked in vegetable stock with wild mushrooms and parmesan.

Spicy Shrimp Pasta — $17.40

Linguine tossed in a garlic-chili tomato sauce with sautéed shrimp.

DESSERTS

Classic Tiramisu — $7.20

Espresso-soaked ladyfingers layered with mascarpone cream and cocoa.

Warm Apple Crumble — $6.80

Baked apples with cinnamon crumble, served with vanilla ice cream.